UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science & Engineering

School of Science & Engineering

PROJECT MANAGEMENT REPORT

UIU CampusConnect

Campus Social & Academic Networking Platform

Vision & Scope • Development Methodology
Risk Management & RMMM • Work Breakdown Structure

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CHAPTER 1

Introduction

1.1 Project Overview

UIU CampusConnect is a university-focused digital platform that brings social networking, academic communication, campus notices, events, clubs, peer-to-peer services and administrative functions into a single shared environment. At present a student at United International University has to move between a notice board, a departmental portal, several unofficial Facebook groups, a handful of WhatsApp or Messenger threads and word-of-mouth in order to keep track of what is happening on campus. Information is duplicated, arrives late, and is frequently impossible to verify. CampusConnect exists to remove that fragmentation.

The platform is built around role-based authentication, personalised feeds, profiles, private messaging, notifications, official notices, academic section announcements, events with registration and QR check-in, club management, course and project groups, a campus marketplace, lost-and-found, global search, moderation and system administration. Every one of these modules is treated as a managed deliverable in this report: it is scoped in Chapter 2, scheduled through the methodology described in Chapter 3, protected by the risk plan in Chapter 4, and decomposed into work packages in Chapter 5.

1.2 Purpose and Structure of this Report

This document is the consolidated project management report for UIU CampusConnect. It is not a design specification and it is not a user manual; it records the management decisions that govern how the system will be built, by whom, in what order, and with what protection against failure. The report is organised as follows.

- **Chapter 2 – Vision and Scope** fixes the boundary of the product. It states what CampusConnect is for, lists the objectives it must achieve, defines the functional areas that are inside the delivery boundary, and explicitly records what the team will *not* build.
- **Chapter 3 – Development Methodology** compares the candidate process models against the actual characteristics of this project and justifies the selection of Agile Scrum, then shows how Scrum ceremonies map onto the CampusConnect modules.
- **Chapter 4 – Risk Management and RMMM** identifies the risks that threaten the platform, defines the team's Risk Mitigation, Monitoring and Management procedure, works three representative risks through that procedure in full, and consolidates all identified risks into a risk plan sheet.
- **Chapter 5 – Work Breakdown Structure** decomposes the entire scope into four hierarchical levels, from the platform itself down to individually assignable work packages, with effort allocations that roll up to 100% of the project.

CHAPTER 2

Vision and Scope

2.1 Vision Statement

> To create a unified digital campus ecosystem that connects students, faculty, university officials and campus organisations through social interaction, academic collaboration, communication and community services.

Each clause of this statement carries a concrete obligation for the CampusConnect build. *Unified* means that a student signs in once and reaches notices, sections, clubs, events and peers from a single feed, rather than maintaining accounts on several disconnected services. *Digital campus ecosystem* means the platform mirrors the real structure of the university – departments, sections, batches, clubs and offices are first-class entities in the data model, not free-text labels. *Connects students, faculty, university officials and campus organisations* is the reason the system is role-based from the very first sprint: a notice published by the Registrar's office must be visibly different from a post written by a classmate, and the platform must be able to prove which is which. Finally, *social interaction, academic collaboration, communication and community services* names the four pillars that later become the top-level branches of the Work Breakdown Structure in Chapter 5.

2.2 Project Objectives

The vision is delivered through six measurable objectives. Each objective below is stated first, then explained in terms of the specific CampusConnect features that satisfy it.

O1. Improve the speed and reach of university communication

Official information at UIU currently travels through physical notice boards, email and informal re-sharing, which means a student who misses one channel misses the announcement entirely. CampusConnect replaces that with the *Official University Notice Board*, where an authorised office publishes once and the notice is targeted to the exact audience it concerns – a department, a batch, a section or the whole campus – and is pushed instantly through the notification centre. Because notices are stored, tagged by priority and searchable, reach is no longer dependent on a student being online at the moment of publication. This objective is realised by WBS elements 4.1.1 and 2.2.2.

O2. Provide a campus-specific social and networking environment

General-purpose social networks mix campus life with unrelated content and admit anyone. CampusConnect restricts registration to verified university email addresses, so every account belongs to a real member of the UIU community. On top of that trusted membership the platform provides the campus feed, posts with rich media, reactions, comments, polls and saved content – the same social affordances students already expect, but scoped to people they actually share a campus with. This objective is realised by WBS elements 2.1.1, 3.1 and 3.2.

O3. Support academic interaction between students and faculty

Academic contact today depends on catching a teacher in the corridor or hoping a reply arrives by email. CampusConnect gives faculty structured profiles carrying office hours, course allocation and research interests, and gives every course section a channel through which the assigned teacher can post announcements that reach exactly the enrolled students. Course, batch and project groups extend the same idea to peer learning, with a shared file area for slides, notes and project documents. This objective is realised by WBS elements 2.1.2, 4.1.2 and 4.3.2.

O4. Centralise events, clubs, notices and community activities

Club recruitment, seminars, workshops and cultural events are currently advertised in whichever group the organiser happens to have access to, and attendance is counted on paper. CampusConnect provides a single events module with publishing, capacity limits, interested/going responses, registration and QR-token check-in, together with a club directory in which every registered club maintains its own profile, membership list, content and gallery. The result is one authoritative campus calendar and one verifiable attendance record. This objective is realised by WBS elements 4.2 and 4.3.1.

O5. Provide peer-to-peer services such as marketplace and lost-and-found

Students routinely resell textbooks, calculators and lab equipment, and lose belongings on campus, but have no trusted place to do either. CampusConnect adds a campus marketplace where verified students list items with photographs, category and price and mark them sold when the exchange is complete, and a lost-and-found register where a finder posts an item and a claimant must supply identifying details before the claim is released. Because both services run on top of verified identities, the platform can attribute every listing and claim to a real student, which is precisely what makes them safer than an anonymous group. This objective is realised by WBS element 5.1.

O6. Maintain role-based security, moderation and administrative control

A platform that carries official notices, private messages and personal profiles cannot be governed informally. CampusConnect enforces a permission matrix in which student, faculty, official and administrator roles each see and do only what their role allows; provides a reporting mechanism and a moderation centre where flagged content is reviewed and actioned; and gives administrators a dashboard with immutable audit logs of every privileged action. This objective is what makes the other five safe to deploy, and it is realised by WBS elements 2.1.1, 5.2 and 1.3.1.

2.3 In-Scope Functional Areas

Table 2.1 defines the delivery boundary of the project. The first two columns state the functional area and the functionality it contains; the third column explains why that area belongs in CampusConnect, so that the scope can be defended when new requests arrive during development.

Table 2.1: In-scope functional areas of UIU CampusConnect and their justification

Area	Included Functionality	Role in the CampusConnect Project
User management	Registration, authentication, role-based permissions, profiles and privacy controls.	The trust anchor of the whole platform. Restricting sign-up to university email domains is what separates CampusConnect from an open social network, and the role assigned at registration decides whether a user may publish an official notice, moderate content or only participate. Nothing else can be built safely until this is in place, so it is the first module scheduled.

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Area	Included Functionality	Role in the CampusConnect Project
Social networking	Campus feed, personalised views, posts, media, reactions, comments, polls and saved content.	The layer that gives students a reason to open the application daily. A personalised feed mixing department, batch and followed sources keeps the platform relevant to each individual, while polls and saved content turn passive reading into participation and let clubs and class representatives gather opinion quickly.
Communication	Connections, following, private messaging and notifications.	Replaces the scattered Messenger and WhatsApp threads students currently use for coursework coordination. Because messaging sits inside a verified campus identity system, a student can be certain who they are talking to, and the central notification service ensures that a notice, an event reminder and a message all reach the user through one consistent channel.
Academic support	Faculty profiles, office hours, research/course information, academic resources and section announcements.	Turns the platform from a social product into an academic one. Section-level announcements guarantee that a change of class time or an assignment deadline reaches every enrolled student and no one else, and published office hours reduce the repeated individual queries that faculty currently receive.
Official communication	Authorised university notices with targeting, priority and filtering.	Gives the university itself a stake in the system. Only accounts with the official role may publish here, which makes the notice board authoritative and prevents the rumour-driven misinformation that circulates in unofficial student groups before examinations and registration.
Events	Event publishing, registration, interest/going actions, capacity and QR/token-based check-in.	Solves a real operational problem for club organisers and the Office of Student Affairs: knowing in advance how many people will attend, enforcing venue capacity, and recording attendance without paper sign-in sheets. QR check-in also produces the participation data clubs need for their annual reports.

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Area	Included Functionality	Role in the CampusConnect Project
Clubs	Club directory, membership requests, club content, events and galleries.	UIU's clubs are a major part of campus life but have no permanent online presence. Giving each club a managed profile with a membership workflow means recruitment, announcements and archives survive the annual change of committee instead of disappearing with a graduating panel.
Groups & resources	Course, batch, project and community groups with shared files.	Supports the collaborative work that dominates a CSE student's semester. A project group with a shared file area and a discussion board keeps code documents, reports and decisions in one place, which is where most team coursework currently breaks down.
Community services	Marketplace and lost-and-found functionality.	The peer-to-peer services of objective O5. They are deliberately in scope because they generate everyday, non-academic reasons to return to the platform, which is what sustains adoption after the novelty of launch.
Search & moderation	Global search, reporting, moderation centre and administrator controls.	The governance layer. Global search makes the accumulated notices, events and listings retrievable rather than merely chronological, while reporting and moderation give the team a defined response path when a user posts abusive content or a fraudulent marketplace listing.

2.4 Out-of-Scope Items

The following capabilities are explicitly excluded from this release. Recording them here is a scope-control measure: it gives the project manager a written basis for refusing late additions, which is the main defence against the scope creep risk registered as R9 in Chapter 4.

- **Full Learning Management System replacement.** CampusConnect carries section announcements and shared group files, but it will not host graded assignment submission, grade books or course content delivery. UIU already operates an LMS for that purpose, and reproducing it would multiply the project's size while duplicating a system the university has no intention of retiring.
- **Online examination and proctoring system.** Examinations demand identity assurance, secure browser control, invigilation and legally defensible audit trails. These requirements are of a different order from social and communication features, and attempting them within a student project timeline would put the delivery of the entire platform at risk.
- **Video-conferencing platform.** Real-time audio and video demands media servers, bandwidth provisioning and continuous operational cost. CampusConnect therefore supports scheduling and announcing an online session and linking out to an existing conferencing tool, rather than transporting the media itself.

- **University-wide ERP or finance system replacement.** Admissions, payroll, accounts and academic records remain in the university's existing administrative systems. CampusConnect may consume identity and section data from them, but it does not become the system of record for institutional data.
- **Direct financial transaction processing and payment gateway services.** The marketplace connects a buyer and a seller and lets them mark an item sold; money changes hands offline. Handling payments would bring PCI-DSS-level compliance obligations, chargeback handling and custody of funds into a student project, which is neither necessary for the objective nor responsibly achievable within it.

2.5 Assumptions and Constraints

The scope above is valid only while the following assumptions hold, and it is bounded by the constraints listed with them.

- **Assumption – institutional email verification is available.** Role-based access depends on being able to verify a `@uiu.ac.bd` or `@bscse.uiu.ac.bd` address at registration. If that check is unavailable, the fake-account risk (R5) rises sharply and a manual approval step must be added.
- **Assumption – section and club data can be obtained.** Section announcements and club directories assume the team can import, or manually seed, an accurate list of sections, courses and registered clubs.
- **Constraint – academic calendar.** The project is executed inside a single semester by a four-member team, which caps total available effort and is the reason the WBS enforces small, individually ownable work packages.
- **Constraint – no operational budget.** Deployment targets free or low-cost hosting tiers, which directly shapes the horizontal-scaling mitigation chosen for the peak-load risk (R3).

CHAPTER 3

Development Methodology

3.1 Selecting a Process Model for CampusConnect

A software methodology is the organised way in which a team moves a product from an idea to a working, maintained system. The choice is not academic: it fixes how requirements are captured, how often working software appears, how change is absorbed and how risk is discovered. For CampusConnect three project characteristics dominate the decision.

1. **The requirements will move.** The platform serves students, faculty, club organisers and administrative offices, and none of these groups can fully specify what they want before they see it. A model that freezes requirements before construction would guarantee that the delivered system is the wrong one.
2. **The system is modular but interdependent.** Ten functional areas can largely be built in parallel, yet they all sit on the same identity, notification and moderation foundations. The process must therefore allow parallel work while forcing frequent integration.
3. **Effort is fixed and small.** Four students working across one semester cannot absorb a heavy documentation overhead, and cannot recover from discovering a fundamental defect late.

3.2 Comparison of Candidate Methodologies

Table 3.1 evaluates the four candidate models against those three characteristics.

Table 3.1: Candidate process models assessed against the needs of UIU CampusConnect

Model	Core Idea	Principal Limitation	Fit for CampusConnect
Waterfall	Strictly sequential phases – requirements, design, implementation, testing, deployment, maintenance – with no return to a completed phase. Produces heavy, complete documentation.	Rigid. Change after sign-off is expensive, and testing happens so late that major defects surface when there is no time to absorb them.	**Rejected.** It would require students, faculty and the administration to agree a complete specification for ten modules before any code exists, and would leave the first working demonstration until the end of the semester.
Prototype	Build a simplified but functional model of the system, show it to users, collect feedback and refine until the requirements are understood.	It is a requirements-elicitation technique, not a complete construction process; it has no built-in answer for scheduling, integration or release.	**Partially adopted.** Prototyping is genuinely useful for the feed and event check-in interfaces, so it is used *inside* sprints as a design activity, not as the overall project model.

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Model	Core Idea	Principal Limitation	Fit for CampusConnect
Spiral	Repeated cycles of planning, risk analysis, engineering and evaluation, with an explicit risk-driven decision at every turn.	Costly and heavy to administer; the ceremony only pays for itself when the consequences of failure are severe.	**Rejected as the overall model.** CampusConnect is risk-aware but not safety-critical or contractually exposed. Its risk discipline is instead supplied by the RMMM process of Chapter 4, executed inside sprints.
Agile Scrum	Iterative and incremental delivery in short sprints from a prioritised product backlog, with defined roles and ceremonies and working software as the measure of progress.	Requires committed, reasonably skilled participation from everyone, and can invite scope creep if the backlog is not controlled.	**Selected.** It matches all three project characteristics: it expects changing requirements, forces integration every sprint, and produces something demonstrable long before the deadline.

3.3 Selected Methodology: Agile Scrum

Agile Scrum is adopted as the development methodology for UIU CampusConnect. The decision rests on four properties of Scrum, each of which answers a specific problem in this project.

Short, timeboxed sprints. Work is delivered in two-week sprints. Within one semester this yields roughly six sprints, so the team reaches a demonstrable, testable increment of CampusConnect within the first month rather than at the end. A defect in the authentication or notification foundation is therefore discovered while there is still time to correct it.

A prioritised product backlog. The ten functional areas are not equally urgent. The backlog lets the team build authentication, profiles and the feed first because everything else depends on them, and defer the marketplace and lost-and-found until the core is stable – without ever pretending that the deferred items have been forgotten.

Frequent stakeholder feedback. Each sprint ends with a demonstration to the supervising faculty member and to sample users drawn from the student body and club committees. Questions such as whether a notice should be filtered by department or by section are answered by people who will actually use the system, in the sprint after they are raised.

Clear ownership through defined roles. With only four members, ambiguity about who owns a module is fatal. Scrum's role structure, combined with the single-point accountability rule applied in the WBS, ensures every work package has exactly one owner.

3.3.1 Scrum Roles in Team 09

3.3.2 Applying the Scrum Cycle to UIU CampusConnect

3.3.3 Indicative Sprint Roadmap

Table 3.4 shows how the WBS modules of Chapter 5 are distributed across the semester. The ordering is driven by dependency: foundations first, then the modules that consume them.

Table 3.2: Allocation of Scrum roles within the CampusConnect team

Role	Responsibility	Application in CampusConnect
Product Owner	Owns and orders the backlog; represents the users.	Collects requirements from students, club representatives and the supervising faculty member, and decides sprint-by-sprint which of the ten modules is next.
Scrum Master	Facilitates ceremonies and removes impediments.	Runs the daily stand-up and the retrospective, and maintains the risk register described in Chapter 4, escalating any risk whose exposure rises.
Development Team	Designs, builds, integrates and tests the increment.	All four members; each takes ownership of specific WBS work packages and is accountable for their integration and testing.

Table 3.3: Scrum activities mapped onto the CampusConnect development cycle

Activity	Application to UIU CampusConnect
Product Backlog	Collect and prioritise platform requirements – authentication, profiles, feed, messaging, notifications, notices, events, clubs, groups, marketplace, lost-and-found, search and moderation – expressed as user stories with acceptance criteria and traced to WBS codes.
Sprint Planning	Select a realistic group of high-priority backlog items for the next two-week sprint, sized against the 8–40 hour work packages defined in the WBS, and agree the sprint goal.
Development Sprint	Design, code, integrate and test the selected features. Every merge runs the automated test suite from WBS 1.2.2 so that integration defects surface within the sprint rather than at the end of the project.
Daily Scrum	A fifteen-minute stand-up covering progress, plan and impediments. New risks raised here are added to the risk register the same day.
Sprint Review	Demonstrate completed functionality to the supervising faculty member and a sample of student users, and record their feedback as new or re-prioritised backlog items.
Sprint Retrospective	Identify what worked, what did not, and one concrete process improvement for the next sprint. Realised risks are reviewed here and their mitigation strategies updated.
Release / Increment	Deploy the stable, tested increment to the staging environment, update the backlog with new learning, and re-baseline the remaining effort against the WBS.

Table 3.4: Indicative sprint roadmap for UIU CampusConnect

Sprint	Primary WBS Coverage	Sprint Goal
1	1.1, 1.2.1	Architecture, data model, UI design system and a working build pipeline.
2	2.1	Verified registration, login, role permissions and student/faculty profiles.
3	3.1, 3.2.1	Campus feed with post creation, media and visibility scoping.
4	2.2, 3.2.2, 3.2.3	Messaging, notification centre, reactions, polls and threaded comments.
5	4.1, 4.2	Official notice board, section announcements, events and QR check-in.
6	4.3, 5.1	Clubs, academic groups, marketplace and lost-and-found.
7	5.2, 1.2.2, 1.3	Moderation, admin dashboard, load testing, security audit and deployment.

CHAPTER 4

Risk Management and RMMM

4.1 The Concept of Risk in this Project

A risk is not a problem yet; it is a possibility of a future problem. Risk analysis and management are the actions a software team takes to understand and control that uncertainty before it becomes damage. For CampusConnect the distinction matters practically: the platform stores personal data for the entire student body, carries official university communication and will face its heaviest load precisely on the days that matter most – registration, section selection and result publication. Discovering these problems in production would be far more expensive than planning for them now.

Every risk in this project is characterised by two quantities, each scored from 0 to 5:

- **Probability** – how likely the risk is to occur during the project or in early operation.
- **Impact** – the severity of the consequence for users, data or the schedule if it does occur.

Their product is the **risk exposure**, and it is the number the team uses to decide where to spend limited effort. Exposure of 20 or above is treated as critical and must have an active mitigation in the current sprint; 10–19 is significant and is scheduled; below 10 is monitored.

4.2 Risk Categories in UIU CampusConnect

Table 4.1: Risk categories and concrete CampusConnect examples

Category	Definition	Examples in CampusConnect
Technical risk	Arising from technology, tools and system design.	The notification service fails to deliver under load; a module's API contract changes and breaks its consumers; the server goes down when the whole batch opens the results notice at once.
Project risk	Arising from planning, scheduling and resource allocation.	A four-member team loses a member to illness mid-sprint; the semester deadline leaves no slack; the backlog keeps growing as reviewers request new features.
Business risk	Threatening the value and adoption of the product.	Students continue to use existing Facebook groups and never adopt the platform; the university declines to endorse the official notice board, removing its authority.
External risk	Originating outside the team's control.	A change in data-protection expectations forces rework of how personal data is stored; hosting provider outage; campus closure disrupting user testing.

4.3 The RMMM Strategy for UIU CampusConnect

RMMM – Risk Mitigation, Monitoring and Management – is the structured plan through which the team handles risk across the project lifecycle. Rather than restating the theory, this section describes the procedure the CampusConnect team actually follows.

4.3.1 Risk Mitigation: acting before the risk occurs

Mitigation is the set of actions taken in advance to reduce either the probability or the impact of a risk. In this project mitigation is not a separate phase; it is embedded in the sprint cycle.

1. **Identification.** At every sprint planning session the team asks what could go wrong with the work packages selected for that sprint. Anything raised is written into the risk register with an owner.
2. **Analysis.** Each risk is scored for probability and impact on the 0–5 scale and its exposure computed. This is what produces the risk plan sheet in Section 4.5.
3. **Preventive design.** Risks with exposure of 20 or above generate concrete engineering work packages in the WBS. Data-privacy risk is why encryption and access-control auditing exist as WBS 1.3.1.1; peak-load risk is why load testing exists as WBS 1.2.2.2; integration risk is why the automated API test suite exists as WBS 1.2.2.1. Mitigation is therefore funded and scheduled, not merely intended.
4. **Assignment.** Every risk has exactly one named owner among the four team members, who is responsible for reporting its status.

4.3.2 Risk Monitoring: watching while the project runs

Monitoring is the continuous process of tracking known risks and detecting new ones. The team's monitoring mechanisms are deliberately lightweight so that they are actually performed.

- The risk register is reviewed at the start of every sprint planning and at every retrospective; probability and impact scores are revised in the light of what the sprint revealed.
- Each critical risk carries a *trigger indicator* – an observable signal that the risk is materialising. Rising API error rates in the staging logs indicate the integration risk; response times climbing above two seconds during load tests indicate the peak-load risk; a falling count of daily active test users indicates the adoption risk.
- The daily stand-up is an explicit opportunity to raise a newly noticed risk, and anything raised there enters the register the same day.
- Sprint velocity is tracked against the WBS effort baseline; a persistent shortfall is itself the trigger indicator for the schedule risk.

4.3.3 Risk Management: responding when the risk occurs

Management covers the actions taken once a risk has actually materialised.

1. **Execute the contingency plan** recorded against that risk, rather than improvising under pressure.
2. **Contain the damage** – restore service, revoke access, roll back the release or restore from backup, whichever the plan specifies.
3. **Communicate** to affected users and to the supervising faculty member, with an honest statement of what happened and what is being done.
4. **Review and improve** at the next retrospective, updating the mitigation so that the same failure cannot recur, and re-scoring the risk.

4.4 Worked RMMM Examples

The three risks below are the highest-exposure entries in the register and are worked through the full RMMM procedure to show how it operates in practice.

RMMM Example 1 — R1: Violation of Data Privacy **Exposure 5 × 5 = 25 (Critical)**

Risk description. CampusConnect stores personal information – email addresses, student and employee IDs, phone numbers, private messages, connection lists and profile media – for the entire user base. Unauthorised disclosure of this data would harm real students and would end the university's willingness to endorse the platform.

Mitigation (before it happens).

- Encrypt data in transit with TLS and encrypt sensitive fields at rest; store passwords only as salted hashes. (WBS 1.3.1.1)
- Apply the role-based permission matrix at the API layer, not only in the interface, so that a crafted request cannot read another user's data. (WBS 2.1.1.3)
- Give students explicit privacy controls over which profile fields are visible to whom. (WBS 2.1.2.1)
- Run a scheduled security audit and penetration test before release. (WBS 1.3.1.2)

Monitoring. Automated access-control tests run on every build; the audit log is reviewed weekly for privileged reads; the trigger indicator is any authorisation-failure spike or any read of a record by an account outside the owning role.

Management (if it happens). Revoke the compromised credentials and invalidate all sessions immediately; isolate the affected service; identify the scope of exposure from the audit log; notify affected users and the department; patch the vulnerability and re-run the full audit before restoring the service.

Owner. Security lead (Development Team).

RMMM Example 2 — R3: Server Down Under Peak Load **Exposure 5 × 5 = 25 (Critical)**

Risk description. Campus traffic is not uniform. During section selection, registration and result publication, a large proportion of the student body opens the platform within the same few minutes. A single under-provisioned server would slow to a halt or fail outright at exactly the moment CampusConnect is most needed, which would permanently damage user confidence in it.

Mitigation (before it happens).

- Design the services to be stateless so that horizontal scaling – several small servers behind a load balancer – is possible instead of relying on one large machine. (WBS 1.1.1.1)
- Cache the feed and notice queries and paginate all list endpoints so that a surge does not translate into unbounded database work. (WBS 3.1.1.3)
- Execute load and peak-traffic tests that simulate the registration-day concurrency before release. (WBS 1.2.2.2)
- Schedule high-volume official notices to publish outside the narrow windows when registration traffic peaks.

Monitoring. Server response time, CPU utilisation and error rate are displayed on the platform health dashboard (WBS 5.2.2.1); the trigger indicator is median response time exceeding two seconds or error rate exceeding 1%.

Management (if it happens). Add server instances behind the load balancer; temporarily disable the most expensive non-essential features, such as the ranked feed, and fall back to a simple chronological view; display an honest status message rather than a blank error; after recovery, re-run the load test at the observed traffic level and raise the baseline capacity.

Owner. DevOps / infrastructure lead (Development Team).

RMMM Example 3 — R8: Module Integration Failure **Exposure 5 × 5 = 25 (Critical)**

Risk description. CampusConnect is assembled from ten interacting modules built in parallel by four people. If the events module and the notification service disagree about a payload, or the moderation module and the feed disagree about what "hidden" means, the platform will not crash — it will quietly show wrong information, such as a notice that never reaches its audience or a removed post that remains visible.

Mitigation (before it happens).

- Define and freeze the interface contract between modules during architecture design, before parallel construction starts. (WBS 1.1.1.1)
- Maintain an automated API and regression test suite that every merge must pass. (WBS 1.2.2.1)
- Integrate continuously through the CI pipeline rather than merging at the end of a sprint. (WBS 1.2.1.1)
- Assign a single technical lead as the arbiter of contract changes, so that no interface changes without both sides being informed.

Monitoring. Build and test results on every commit; the trigger indicator is any failing contract test or a rise in integration-level defects reported during a sprint review.

Management (if it happens). Revert the offending merge to restore a known-good build; reproduce the mismatch with a failing test; correct the contract on both sides and add a permanent regression test; record the incident in the retrospective so the contract review step is strengthened.

Owner. Technical lead (Development Team).

4.5 Risk Plan Sheet

A risk plan sheet is the consolidated list of all risks that threaten the project together with the strategy for mitigating each. Table 4.2 records the ten risks identified for UIU CampusConnect. Probability and impact are scored from 0 to 5 and priority is their product.

Table 4.2: Risk plan sheet for UIU CampusConnect (probability and impact scored 0–5)

Risk ID	Risk Description	Prob.	Impact	Priority	Mitigation Strategy	Owner
R1	**Violation of data privacy.** The system stores personal information such as email, student/employee ID, phone number, messages and friend lists.	5	5	25	End-to-end encryption; enforce access control at the API layer; regular security testing and audit.	Security lead
R2	**Unauthorized access.** A user gains the permissions of another role, for example a student publishing an official notice.	4	5	20	Role-based access control with a documented permission matrix; server-side authorisation validation on every endpoint.	Developer team
R3	**Server down.** During section selection, registration and result publication, many users access the server simultaneously and it fails.	5	5	25	Horizontal scaling behind a load balancer; caching and pagination; load testing before release.	DevOps engineer
R4	**Marketplace fraud.** A user creates a fake listing or provides misleading information about an item.	3	4	12	Seller verification against university identity; reporting mechanism; sold-status controls and listing expiry.	Marketplace manager

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Risk ID	Risk Description	Prob.	Impact	Priority	Mitigation Strategy	Owner
R5	**Fake account creation.** Someone from outside the university registers and enters the system.	4	4	16	Restrict registration to the university's official email domain with mandatory verification; manual review of anomalies.	IT admin
R6	**Lost-and-found abuse.** Someone falsely claims that a found item belongs to them.	3	3	9	Require identifying item details before a claim is released; allow users to report false claims; log all claims.	Community manager
R7	**User reluctance.** After the system is built, students show no interest in switching from existing groups.	3	3	9	Promote the platform before launch through clubs and class representatives; user-friendly design validated by sprint reviews with real students.	Product owner
R8	**Module integration failure.** One module does not communicate correctly with another and the system produces incorrect information.	5	5	25	Clearly defined interfaces frozen at design time; automated API and regression testing on every merge; continuous integration.	Technical lead
R9	**Scope creep.** The project has many features and the team keeps adding more, so the work grows beyond the available effort.	4	5	20	Fixed feature baseline recorded in the Vision & Scope chapter; formal approval required for any new requirement; out-of-scope list enforced.	Project manager
R10	**Data loss.** The system stores a large volume of information and server failure could destroy it.	4	4	16	Scheduled automated backups with periodic restore drills; database replication; retention of audit logs.	Database administrator

4.6 Interpreting the Risk Profile

Four risks – R1, R3, R8 and R9 – carry an exposure of 20 or above and are therefore critical. It is worth noting what they have in common: three of them (R1, R3, R8) are consequences of the platform's own ambition. CampusConnect is valuable precisely because it centralises identity, traffic and modules, and centralisation is exactly what makes privacy breaches, load failures and integration defects damaging. The fourth, R9, is a consequence of the same breadth seen from the management side.

This is why the mitigations for these four are not advisory notes but funded elements of the Work Breakdown Structure in Chapter 5: security auditing (1.3.1), load testing (1.2.2.2), automated contract testing (1.2.2.1) and a fixed, documented scope baseline. The remaining six risks are real but either lower in probability or recoverable, and are managed through monitoring and the standard sprint response.

CHAPTER 5

Work Breakdown Structure

5.1 Introduction

A Work Breakdown Structure (WBS) is a deliverable-oriented, hierarchical decomposition of the total scope of work to be carried out by the team in order to achieve the project objectives and produce the required deliverables. It is the bridge between the scope defined in Chapter 2 and the sprint plan of Chapter 3: the scope says *what* the platform is, and the WBS says *what pieces of work* produce it.

For UIU CampusConnect – an integrated campus networking hub combining social feeds, academic section alerts, official notice dissemination, event check-in tools, club administration, group collaboration and peer-to-peer services – the WBS performs three jobs. It fixes the boundary of the system in work terms, so that anything not appearing in the structure is by definition out of scope. It partitions intricate functional areas into small, manageable work packages that one of the four team members can own outright. And it supplies the baseline against which scheduling, effort estimation, sprint sizing, risk monitoring and progress reporting are all performed.

5.2 Principles Applied

The CampusConnect WBS is constructed on the following project management principles.

1. **Total scope capture (the 100% rule).** The structure accounts for 100% of the project scope, including internal, interim and external deliverables. The effort content of every set of child elements sums exactly to that of its parent, with no omissions and no additions. Section 5.7 demonstrates this arithmetically for every branch.

2. **Distinct work boundaries (mutually exclusive elements).** Every work package is uniquely defined with non-overlapping boundaries. This matters concretely in CampusConnect because several modules touch the same data: the notice board, the events module and the messaging module all raise notifications, so the notification engine is placed once, at 2.2.2.1, and consumed by the others rather than being rebuilt three times.

3. **Deliverable-focused structure.** The hierarchy is organised around tangible system outcomes expressed as noun phrases – "QR/token generation and scan-based check-in", not "write the check-in code". This keeps the structure stable even when the implementation approach changes inside a sprint.

4. **Four-level decomposition.** The structure is developed to four numbered levels. Level 1 is the platform, Level 2 the five major system modules, Level 3 the deliverable components inside each module, and Level 4 the terminal work packages that are actually assigned, estimated and tracked. The original three-level structure stopped at component granularity, which was too coarse to schedule against two-week sprints; the fourth level closes that gap.

5. **Optimal granularity (the 40-hour rule).** Level-4 work packages are sized at roughly 8 to 40 direct labour hours – one to five person-days – which is small enough to track meaningfully within a two-week sprint and large enough to avoid micromanagement. With four members over a semester the project baseline is approximately 1,000 person-hours, so one percentage point of effort corresponds to about ten hours; the level-4 packages, sized at 1% to 2%, therefore fall inside the 8–40 hour band by construction.

6. **Balanced work sizing.** Level-3 deliverables carry roughly 3% to 6% of total effort each, which keeps effort evenly distributed across the functional modules and prevents any single component from becoming an unmanageable block. The level-4 packages beneath them are correspondingly sized at 1% to 2%.

5.3 Rules Followed

1. **Cross-functional collaboration.** The breakdown was formulated jointly by all four team members so that inter-module dependencies – feed on identity, notices on notifications, moderation on posts – were respected rather than discovered later.

2. **Systematic identification and numbering.** A standard decimal coding system (1.0, 1.1, 1.1.1, 1.1.1.1) is assigned to every element. These codes are used as the cross-reference key throughout this report: backlog items in Chapter 3 and mitigation actions in Chapter 4 both cite WBS codes, so a mitigation can always be traced to funded work.

3. **Iterative decomposition.** Each level refines the level above it into more granular, well-defined components, stopping as soon as an element is small enough to be estimated confidently and owned by one person.

4. **Single-point accountability.** Every terminal work package is assigned to exactly one owner, accountable for its execution, its quality and its delivery. Shared ownership is not permitted at Level 4.

5.4 Level Structure

Table 5.1: Meaning of each WBS level in UIU CampusConnect

Level	Element Type	Example	Management Use
1	Project / platform	UIU CampusConnect	Defines the 100% baseline.
2	System module	2.0 Identity & Social Network	Owned by a module lead; maps to a release theme.
3	Deliverable component	2.1.1 Role-Based Auth & Access Control	Sized at 3–6%; becomes an epic in the backlog.
4	Work package	2.1.1.1 UIU email-domain registration & verification	Sized at 8–40 hours; assigned to one person and tracked in the sprint.

5.5 Level 1 and Level 2 Overview

At the highest level the platform divides into five modules. Their effort allocations sum to 100% of the project.

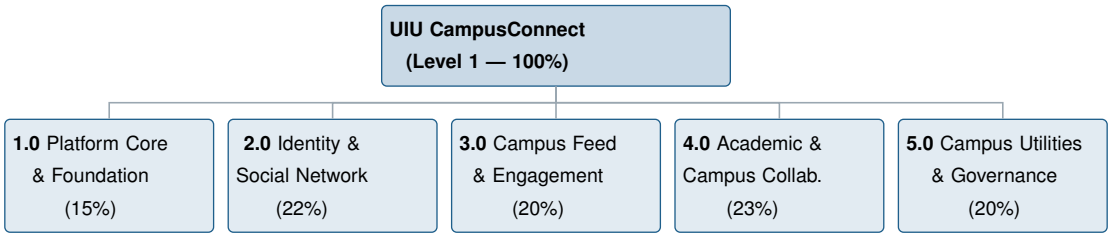


Figure 5.1: Level 1 and Level 2 of the UIU CampusConnect Work Breakdown Structure

Table 5.2: Level-2 modules and their contribution to the project

Code	Module	Effort	Why it carries this share of the work
1.0	Platform Core & Foundation	15%	Architecture, design system, build pipeline, testing infrastructure, security audit and deployment. It produces no user-visible feature, but every other module depends on it and three of the four critical risks are mitigated here.
2.0	Identity & Social Network	22%	Verified registration, roles, profiles, the campus directory, private messaging and the notification engine. Large because it is the trust and communication foundation that objectives O2, O3 and O6 all rest on.
3.0	Campus Feed & Engagement	20%	The feed engine, ranking, global search, post creation, reactions, polls, bookmarks and threaded comments — the daily-use surface of the platform.
4.0	Academic & Campus Collaboration	23%	Official notices, section announcements, events with QR check-in, clubs and academic groups. The largest module because it delivers objectives O1, O3 and O4 and is what distinguishes CampusConnect from a generic social network.
5.0	Campus Utilities & Governance	20%	Marketplace, lost-and-found, reporting, moderation, the admin dashboard and audit logging — objective O5 plus the governance half of O6.
	Total	**100%**	

5.6 Four-Level Decomposition

The five subsections below give the complete four-level structure. Each module is shown as a tree with effort percentages, followed by a dictionary of its Level-4 work packages stating the deliverable that completes each one.

> **Note on corrections to the original allocation.** Two arithmetic inconsistencies in the earlier three-level draft were corrected so that the 100% rule holds at every node. First, the children of 3.2 previously summed to 13% against an 11% parent and have been rebalanced to 5%, 4% and 2%. Second, element 5.1 was shown as 20% although its children summed to 10% and its parent 5.0 totals 20%; it is corrected to 10%, and the two components of 5.2 are allocated 5% each. All Level-2 totals are unchanged.

5.6.1 1.0 Platform Core & Foundation (15%)

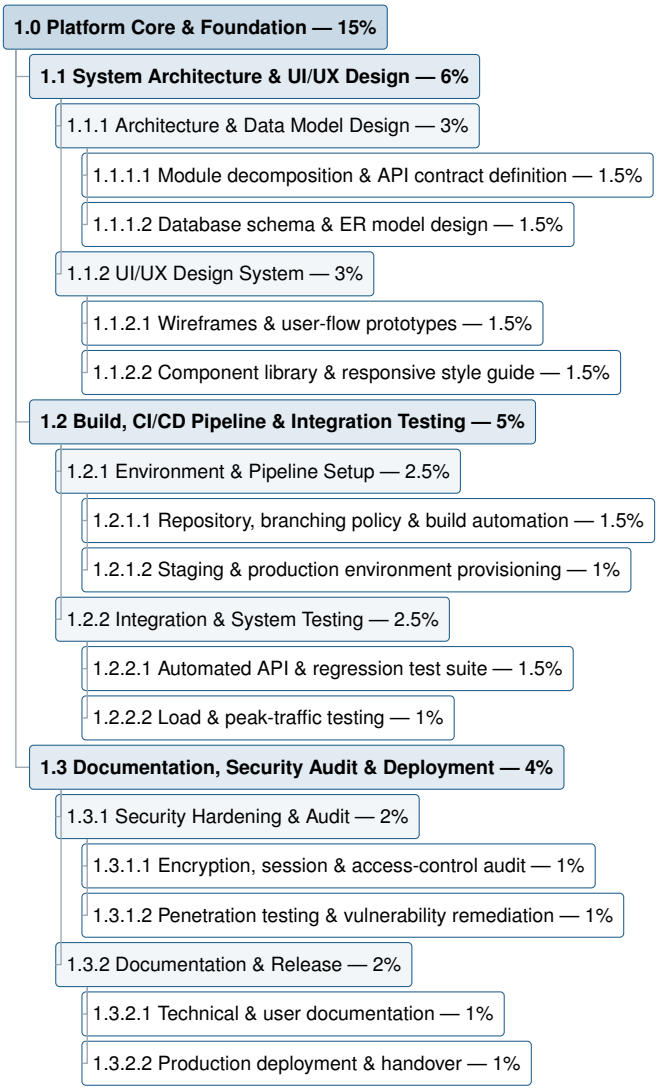


Figure 5.2: Four-level decomposition of Module 1.0 — Platform Core & Foundation

Table 5.3: Level-4 work package dictionary for Module 1.0

Code	Work Package	Completion Deliverable
1.1.1.1	Module decomposition & API contract definition	Signed-off interface specification for all ten modules; the baseline that mitigates risk R8.
1.1.1.2	Database schema & ER model design	Normalised schema covering users, roles, posts, notices, events, clubs, groups and listings.
1.1.2.1	Wireframes & user-flow prototypes	Clickable prototype of the primary flows, validated in a sprint review with student users.
1.1.2.2	Component library & responsive style guide	Reusable UI component set rendering correctly on mobile and desktop.
1.2.1.1	Repository, branching policy & build automation	Version-controlled repository with a pipeline that builds and tests every merge.
1.2.1.2	Staging & production environment provisioning	Two reproducible environments with configuration separated from code.
1.2.2.1	Automated API & regression test suite	Contract tests for every inter-module interface, run on each commit.
1.2.2.2	Load & peak-traffic testing	Test report demonstrating acceptable response times at registration-day concurrency; mitigation for risk R3.
1.3.1.1	Encryption, session & access-control audit	Audit record confirming TLS, hashed credentials and server-side authorisation on every endpoint.
1.3.1.2	Penetration testing & vulnerability remediation	Findings log with all critical and high issues closed; mitigation for risk R1.
1.3.2.1	Technical & user documentation	Deployment guide, API reference and end-user help content.
1.3.2.2	Production deployment & handover	Platform running in production with monitoring enabled and handover completed.

5.6.2 2.0 Identity & Social Network (22%)

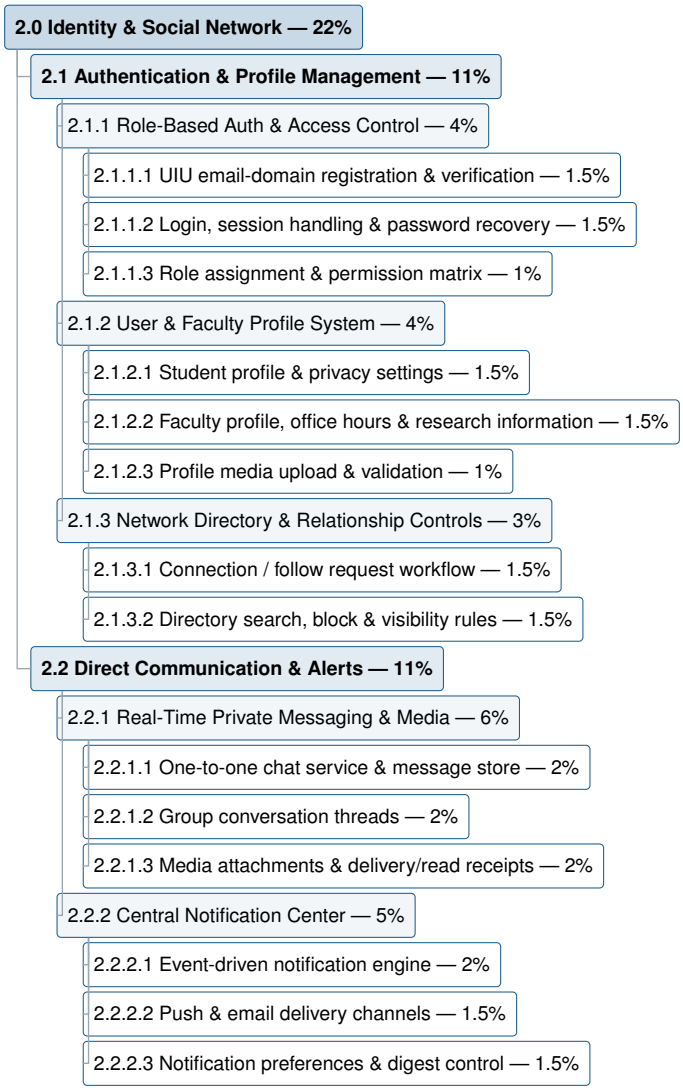


Figure 5.3: Four-level decomposition of Module 2.0 — Identity & Social Network

Table 5.4: Level-4 work package dictionary for Module 2.0

Code	Work Package	Completion Deliverable
2.1.1.1	UIU email-domain registration & verification	Sign-up restricted to university domains with mandatory email confirmation; mitigation for risk R5.
2.1.1.2	Login, session handling & password recovery	Secure authentication with expiring sessions and a verified reset flow.
2.1.1.3	Role assignment & permission matrix	Documented and enforced matrix for student, faculty, official and administrator roles; mitigation for risk R2.
2.1.2.1	Student profile & privacy settings	Editable profile with per-field visibility control.
2.1.2.2	Faculty profile, office hours & research information	Faculty-specific profile supporting objective O3.
2.1.2.3	Profile media upload & validation	Validated image upload with size and type restrictions.
2.1.3.1	Connection / follow request workflow	Request, accept, decline and unfollow operations with notifications.
2.1.3.2	Directory search, block & visibility rules	Campus directory honouring privacy settings and block lists.
2.2.1.1	One-to-one chat service & message store	Persistent private messaging between two verified users.
2.2.1.2	Group conversation threads	Multi-participant threads with membership management.
2.2.1.3	Media attachments & delivery/read receipts	File and image sharing in chat with delivery and read status.
2.2.2.1	Event-driven notification engine	Single shared service consumed by notices, events, clubs and messaging.
2.2.2.2	Push & email delivery channels	Notifications delivered outside the application, supporting objective O1.
2.2.2.3	Notification preferences & digest control	Per-category user preferences and a digest option to prevent alert fatigue.

5.6.3 3.0 Campus Feed & Engagement (20%)

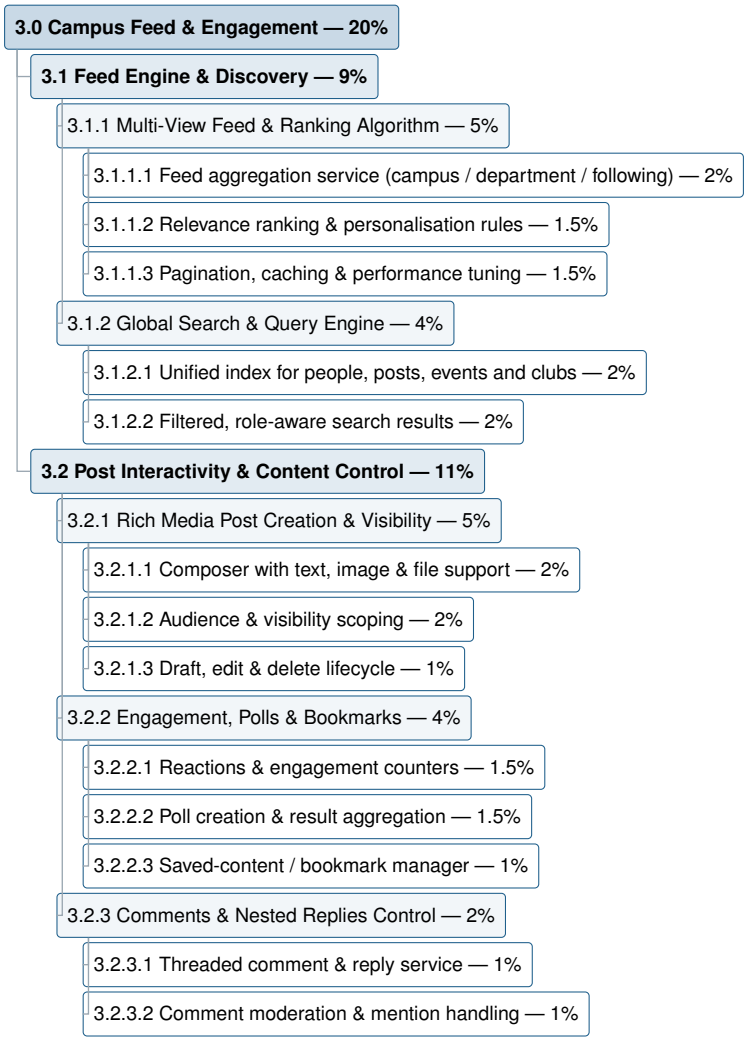


Figure 5.4: Four-level decomposition of Module 3.0 — Campus Feed & Engagement

Table 5.5: Level-4 work package dictionary for Module 3.0

Code	Work Package	Completion Deliverable
3.1.1.1	Feed aggregation service	Feed assembled from campus-wide, departmental and followed sources.
3.1.1.2	Relevance ranking & personalisation rules	Ranking that surfaces a student's own section, batch and clubs first.
3.1.1.3	Pagination, caching & performance tuning	Bounded query cost per request; mitigation for risk R3.
3.1.2.1	Unified search index	Single index spanning people, posts, notices, events and clubs.
3.1.2.2	Filtered, role-aware search results	Results filtered by type and restricted to what the requesting role may see.
3.2.1.1	Composer with text, image & file support	Post composer with validated media upload.
3.2.1.2	Audience & visibility scoping	Author-selected audience: public, department, batch, section or group.
3.2.1.3	Draft, edit & delete lifecycle	Full post lifecycle with edit history retained for moderation.
3.2.2.1	Reactions & engagement counters	Reaction set with accurate aggregate counts.
3.2.2.2	Poll creation & result aggregation	Polls with single or multiple choice and live results.
3.2.2.3	Saved-content / bookmark manager	Personal saved list retrievable across sessions.
3.2.3.1	Threaded comment & reply service	Nested replies with correct ordering and pagination.
3.2.3.2	Comment moderation & mention handling	Mentions raising notifications; comments reportable to the moderation queue.

5.6.4 4.0 Academic & Campus Collaboration (23%)

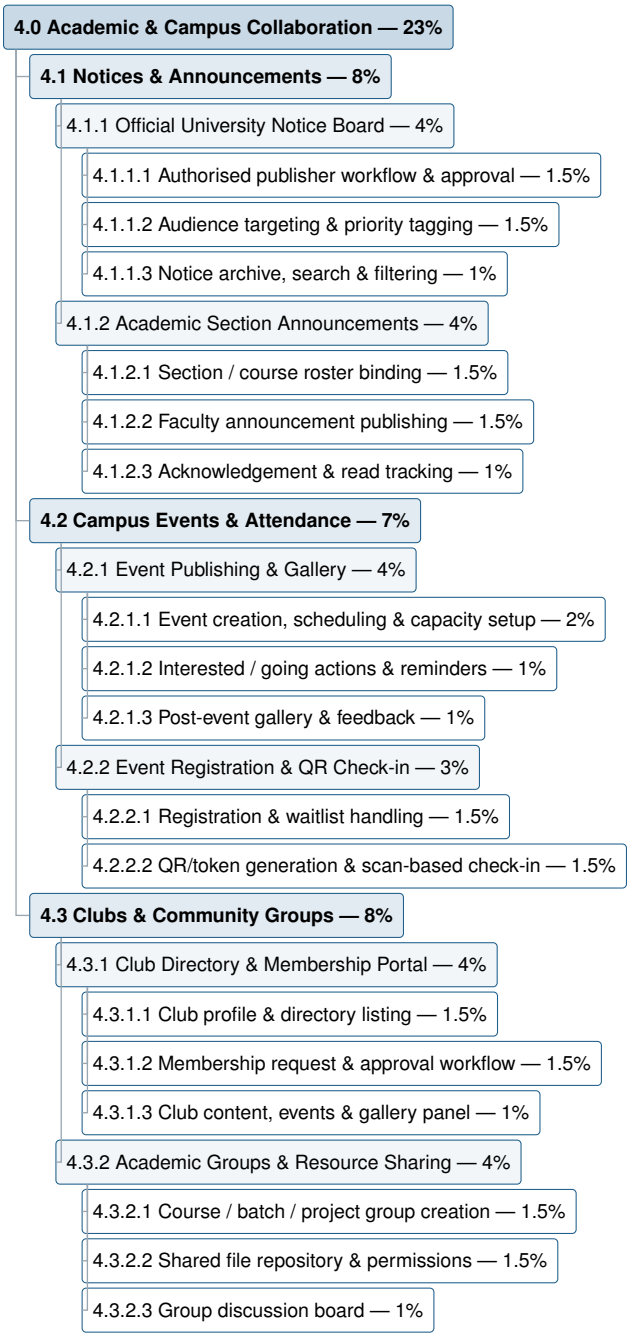


Figure 5.5: Four-level decomposition of Module 4.0 — Academic & Campus Collaboration

Table 5.6: Level-4 work package dictionary for Module 4.0

Code	Work Package	Completion Deliverable
4.1.1.1	Authorised publisher workflow & approval	Only accounts with the official role may publish; approval trail recorded.
4.1.1.2	Audience targeting & priority tagging	Notices targeted to campus, department, batch or section, with priority levels.
4.1.1.3	Notice archive, search & filtering	Full notice history retrievable and filterable, supporting objective O1.
4.1.2.1	Section / course roster binding	Sections linked to enrolled students and the assigned faculty member.
4.1.2.2	Faculty announcement publishing	Section announcements reaching exactly the enrolled students.
4.1.2.3	Acknowledgement & read tracking	Faculty view of who has seen a time-critical announcement.
4.2.1.1	Event creation, scheduling & capacity setup	Event records with venue, schedule and enforced capacity limits.
4.2.1.2	Interested / going actions & reminders	Attendance intent captured and reminders raised through the notification engine.
4.2.1.3	Post-event gallery & feedback	Photo gallery and feedback collection for organiser reporting.
4.2.2.1	Registration & waitlist handling	Registration with automatic waitlist promotion when capacity frees.
4.2.2.2	QR/token generation & scan-based check-in	Unique token per registration and a verified check-in record.
4.3.1.1	Club profile & directory listing	Searchable directory of registered clubs with managed profiles.
4.3.1.2	Membership request & approval workflow	Join requests reviewed by the club committee with roster maintenance.
4.3.1.3	Club content, events & gallery panel	Club-scoped posts, events and media archive surviving committee changes.
4.3.2.1	Course / batch / project group creation	Group types matching the real academic structure.
4.3.2.2	Shared file repository & permissions	Group file area with upload, versioned replacement and access control.
4.3.2.3	Group discussion board	Threaded discussion within each group, supporting objective O3.

5.6.5 5.0 Campus Utilities & Governance (20%)

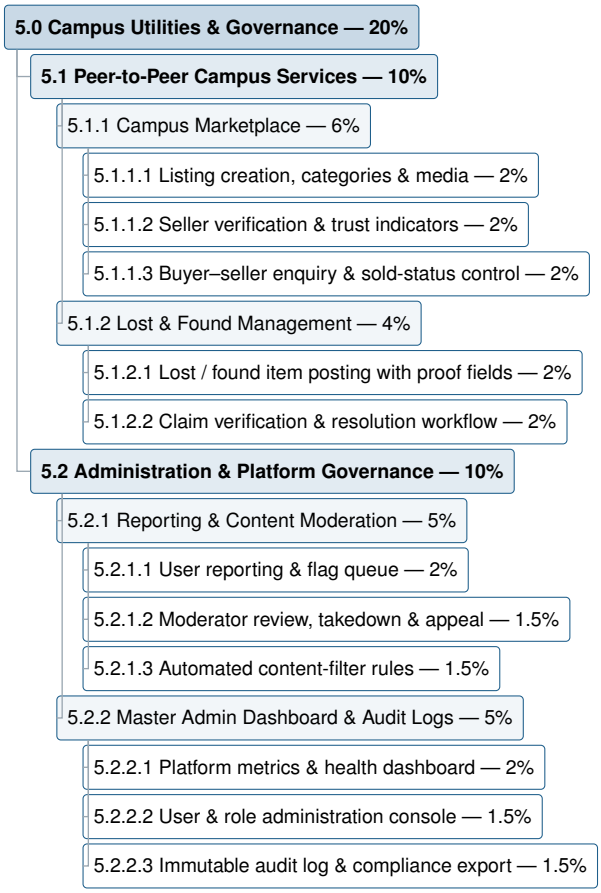


Figure 5.6: Four-level decomposition of Module 5.0 — Campus Utilities & Governance

Table 5.7: Level-4 work package dictionary for Module 5.0

Code	Work Package	Completion Deliverable
5.1.1.1	Listing creation, categories & media	Item listings with category, price, condition and photographs.
5.1.1.2	Seller verification & trust indicators	Every listing attributed to a verified student, with report counts visible; mitigation for risk R4.
5.1.1.3	Buyer–seller enquiry & sold-status control	Enquiry channel and mandatory sold marking with listing expiry.
5.1.2.1	Lost / found item posting with proof fields	Item posts that deliberately withhold one identifying detail for claim verification.
5.1.2.2	Claim verification & resolution workflow	Claimant must supply the withheld detail before release; mitigation for risk R6.
5.2.1.1	User reporting & flag queue	Report action on posts, comments, listings and profiles feeding one queue.
5.2.1.2	Moderator review, takedown & appeal	Defined review path with recorded decisions and an appeal route.
5.2.1.3	Automated content-filter rules	Rule-based pre-screening that flags obvious abuse before a human review.
5.2.2.1	Platform metrics & health dashboard	Live view of active users, response time and error rate; the monitoring surface for risk R3.
5.2.2.2	User & role administration console	Administrator control over accounts, roles and suspensions.
5.2.2.3	Immutable audit log & compliance export	Tamper-evident record of every privileged action; the evidence base for investigating risk R1.

5.7 Effort Roll-Up Verification

Table 5.8 demonstrates that the 100% rule holds at every node of the structure: the sum of each element's children equals the element itself, and the five Level-2 modules total 100%.

Table 5.8: Verification of effort roll-up across all four WBS levels

Parent Element	Parent %	Children (%)	Sum
1.0 Platform Core	15	1.1 (6) + 1.2 (5) + 1.3 (4)	15 ✓
1.1	6	1.1.1 (3) + 1.1.2 (3)	6 ✓
1.1.1	3	1.5 + 1.5	3 ✓
1.1.2	3	1.5 + 1.5	3 ✓
1.2	5	1.2.1 (2.5) + 1.2.2 (2.5)	5 ✓
1.2.1 / 1.2.2	2.5 each	1.5 + 1 (each)	2.5 ✓
1.3	4	1.3.1 (2) + 1.3.2 (2)	4 ✓
1.3.1 / 1.3.2	2 each	1 + 1 (each)	2 ✓
2.0 Identity & Social	22	2.1 (11) + 2.2 (11)	22 ✓
2.1	11	2.1.1 (4) + 2.1.2 (4) + 2.1.3 (3)	11 ✓
2.1.1 / 2.1.2	4 each	1.5 + 1.5 + 1 (each)	4 ✓
2.1.3	3	1.5 + 1.5	3 ✓
2.2	11	2.2.1 (6) + 2.2.2 (5)	11 ✓
2.2.1	6	2 + 2 + 2	6 ✓
2.2.2	5	2 + 1.5 + 1.5	5 ✓
3.0 Feed & Engagement	20	3.1 (9) + 3.2 (11)	20 ✓
3.1	9	3.1.1 (5) + 3.1.2 (4)	9 ✓
3.1.1	5	2 + 1.5 + 1.5	5 ✓
3.1.2	4	2 + 2	4 ✓
3.2	11	3.2.1 (5) + 3.2.2 (4) + 3.2.3 (2)	11 ✓
3.2.1	5	2 + 2 + 1	5 ✓
3.2.2	4	1.5 + 1.5 + 1	4 ✓
3.2.3	2	1 + 1	2 ✓
4.0 Academic & Collab.	23	4.1 (8) + 4.2 (7) + 4.3 (8)	23 ✓
4.1	8	4.1.1 (4) + 4.1.2 (4)	8 ✓
4.1.1 / 4.1.2	4 each	1.5 + 1.5 + 1 (each)	4 ✓
4.2	7	4.2.1 (4) + 4.2.2 (3)	7 ✓
4.2.1	4	2 + 1 + 1	4 ✓
4.2.2	3	1.5 + 1.5	3 ✓
4.3	8	4.3.1 (4) + 4.3.2 (4)	8 ✓
4.3.1 / 4.3.2	4 each	1.5 + 1.5 + 1 (each)	4 ✓
5.0 Utilities & Governance	20	5.1 (10) + 5.2 (10)	20 ✓
5.1	10	5.1.1 (6) + 5.1.2 (4)	10 ✓
5.1.1	6	2 + 2 + 2	6 ✓
5.1.2	4	2 + 2	4 ✓
5.2	10	5.2.1 (5) + 5.2.2 (5)	10 ✓
5.2.1 / 5.2.2	5 each	2 + 1.5 + 1.5 (each)	5 ✓
Project total (Level 2)		**15 + 22 + 20 + 23 + 20**	**100 ✓**

CHAPTER 6

Conclusion

This report has set out the management baseline for UIU CampusConnect. The Vision and Scope chapter fixed what the platform is for and, equally importantly, what it is not: ten functional areas were admitted into the delivery boundary with a stated reason for each, and five capabilities were formally excluded so that the team has written grounds for refusing them later. The methodology chapter compared the candidate process models against the three defining characteristics of this project – moving requirements, modular but interdependent construction, and a small fixed effort budget – and selected Agile Scrum, mapping its ceremonies onto a seven-sprint semester roadmap.

The risk chapter converted risk from a list of worries into a working procedure. Ten risks were scored for probability and impact, four were found to be critical, and the RMMM discipline was described as the team actually operates it: risks identified at sprint planning, mitigations funded as WBS work packages, trigger indicators watched during the sprint, and a defined contingency executed if a risk materialises. Three of those risks were worked through the full cycle to demonstrate the procedure.

Finally, the Work Breakdown Structure decomposed the entire scope into four levels, from the platform down to sixty-seven terminal work packages, each sized for a two-week sprint and each assignable to a single owner. The effort allocations were corrected where the earlier draft did not balance, and the roll-up was verified to 100% at every node.

Taken together, these four chapters form a single coherent plan: every objective in Chapter 2 is traceable to work packages in Chapter 5, every critical risk in Chapter 4 is mitigated by a funded element of that same structure, and every element is scheduled through the sprint cycle of Chapter 3. That traceability is what makes the plan executable rather than merely descriptive.